

## BAMAKO, Mali

WMO: 612910

Lat: 12.534N

Lon: 7.950W

Elev: 380

StdP: 96.84

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 15.0       | 16.2 | -10.2                         | 1.6 | 30.5 | -8.2 | 2.0 | 27.0 | 8.6                   | 25.8 | 7.9 | 26.4 | 1.9                   | 20   | 0.376 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 13.3                   | 40.5            | 19.7 | 39.8 | 19.7 | 38.9 | 19.9 | 27.0                | 31.4 | 26.2 | 30.8 | 25.8 | 30.4 | 3.6                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.1                            | 22.5 | 28.5 | 25.2 | 21.3 | 27.6 | 24.9 | 21.0 | 27.4 | 87.8          | 31.0 | 84.3 | 31.0 | 82.4 | 30.5 | 36.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |     |            |     |            |     |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-----|------------|-----|------------|-----|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |     | n=20 years |     | n=50 years |     |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max | Min        | Max | Min        | Max |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |  |
| 7.9               | 6.9  | 6.1 | DB                         | 11.6 | 42.1               | 2.0 | 0.7                                                | 10.2 | 42.6       | 9.0 | 43.1       | 7.9 | 43.5       | 6.5 | 44.0 |  |
|                   |      |     | WB                         | 5.7  | 28.9               | 1.3 | 1.9                                                | 4.8  | 30.2       | 4.0 | 31.4       | 3.2 | 32.4       | 2.2 | 33.8 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 28.0   | 24.9  | 27.8  | 30.8  | 32.4  | 31.6  | 29.2  | 26.9  | 26.2  | 26.8  | 28.0  | 26.7  | 25.0  |
|                                                                        | DBStd       | 2.98   | 2.10  | 2.36  | 1.92  | 1.70  | 2.04  | 2.08  | 1.65  | 1.34  | 1.31  | 1.15  | 1.40  | 1.80  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD10.0     | 6578   | 461   | 498   | 644   | 673   | 671   | 576   | 523   | 502   | 505   | 558   | 501   | 465   |
|                                                                        | CDD18.3     | 3536   | 203   | 265   | 386   | 423   | 412   | 326   | 265   | 244   | 255   | 299   | 251   | 207   |
|                                                                        | CDH23.3     | 42842  | 2163  | 3316  | 5631  | 6579  | 5957  | 4012  | 2492  | 1903  | 2234  | 3169  | 3085  | 2301  |
|                                                                        | CDH26.7     | 23675  | 1066  | 1878  | 3509  | 4309  | 3667  | 2098  | 982   | 639   | 917   | 1653  | 1779  | 1178  |
| (14) Wind                                                              | WSAvg       | 2.8    | 3.5   | 3.6   | 3.2   | 3.0   | 3.2   | 3.1   | 2.5   | 2.3   | 2.0   | 1.9   | 2.5   | 3.3   |
| (15) Precipitation                                                     | PrecAvg     | 944    | 0     | 0     | 4     | 21    | 58    | 121   | 230   | 264   | 188   | 67    | 3     | 0     |
|                                                                        | PrecMax     | 1492   | 8     | 4     | 34    | 137   | 143   | 286   | 409   | 467   | 502   | 256   | 30    | 25    |
|                                                                        | PrecMin     | 643    | 0     | 0     | 0     | 0     | 3     | 39    | 108   | 129   | 87    | 0     | 0     | 0     |
|                                                                        | PrecStd     | 174    | 1     | 1     | 8     | 26    | 36    | 45    | 68    | 77    | 75    | 55    | 7     | 3     |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.0  | 39.2  | 40.9  | 41.9  | 41.2  | 39.0  | 35.1  | 33.1  | 34.9  | 37.0  | 37.0  | 36.1  |
|                                                                        |             | MCWB   | 16.6  | 17.5  | 18.6  | 19.2  | 20.6  | 22.2  | 24.6  | 25.5  | 25.5  | 21.7  | 19.0  | 17.5  |
|                                                                        | 2%          | DB     | 35.0  | 38.0  | 39.9  | 40.9  | 40.1  | 37.2  | 33.8  | 32.1  | 33.8  | 35.9  | 36.1  | 35.0  |
|                                                                        |             | MCWB   | 16.3  | 17.1  | 18.0  | 19.5  | 21.2  | 23.6  | 24.8  | 25.2  | 25.2  | 22.1  | 18.9  | 17.0  |
|                                                                        | 5%          | DB     | 33.8  | 36.9  | 39.0  | 40.0  | 39.0  | 36.0  | 32.5  | 31.1  | 32.5  | 35.0  | 35.5  | 34.0  |
|                                                                        |             | MCWB   | 15.6  | 16.9  | 18.1  | 19.8  | 22.0  | 24.0  | 24.7  | 24.9  | 25.1  | 23.0  | 18.8  | 16.5  |
|                                                                        | 10%         | DB     | 32.2  | 35.7  | 38.0  | 39.1  | 37.9  | 34.8  | 31.3  | 30.1  | 31.4  | 34.0  | 34.8  | 32.8  |
|                                                                        |             | MCWB   | 15.0  | 16.3  | 17.8  | 20.1  | 22.3  | 24.2  | 24.6  | 24.8  | 25.0  | 23.5  | 18.6  | 16.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 20.0  | 22.3  | 24.0  | 26.5  | 27.1  | 27.2  | 27.4  | 27.5  | 27.8  | 27.4  | 24.7  | 20.0  |
|                                                                        |             | MCDB   | 28.3  | 32.3  | 34.6  | 34.0  | 33.5  | 32.9  | 30.7  | 30.2  | 31.4  | 31.6  | 30.5  | 30.8  |
|                                                                        | 2%          | WB     | 18.2  | 20.2  | 22.8  | 25.1  | 25.9  | 26.2  | 26.2  | 26.2  | 26.6  | 26.2  | 23.5  | 18.6  |
|                                                                        |             | MCDB   | 29.8  | 32.5  | 33.0  | 33.0  | 32.6  | 31.9  | 30.1  | 29.5  | 30.3  | 31.0  | 29.5  | 31.7  |
|                                                                        | 5%          | WB     | 16.9  | 18.3  | 21.8  | 24.2  | 25.3  | 25.7  | 25.6  | 25.7  | 25.9  | 25.5  | 22.5  | 17.7  |
|                                                                        |             | MCDB   | 30.7  | 33.4  | 32.0  | 33.1  | 32.1  | 31.4  | 29.5  | 29.0  | 29.8  | 30.4  | 28.5  | 31.3  |
|                                                                        | 10%         | WB     | 15.8  | 17.0  | 20.8  | 23.4  | 24.7  | 25.2  | 25.2  | 25.2  | 25.4  | 25.1  | 21.4  | 16.7  |
|                                                                        |             | MCDB   | 30.6  | 33.6  | 32.3  | 32.6  | 31.8  | 30.8  | 29.1  | 28.4  | 29.3  | 29.8  | 28.2  | 30.5  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.0  | 15.0  | 14.5  | 13.3  | 11.9  | 10.6  | 8.8   | 8.2   | 9.6   | 12.5  | 16.7  | 15.8  |
|                                                                        |             | MCDBR  | 16.9  | 16.4  | 15.5  | 14.8  | 13.5  | 12.3  | 10.3  | 9.6   | 11.1  | 14.3  | 17.9  | 17.7  |
|                                                                        |             | MCWBR  | 6.7   | 6.3   | 6.0   | 5.8   | 4.6   | 3.9   | 3.8   | 4.0   | 4.5   | 4.7   | 5.7   | 6.0   |
|                                                                        | 5% WB       | MCDBR  | 15.5  | 14.7  | 12.9  | 12.1  | 11.6  | 10.9  | 9.4   | 8.9   | 10.2  | 12.0  | 14.7  | 16.0  |
|                                                                        |             | MCWBR  | 7.1   | 6.8   | 6.5   | 5.4   | 4.2   | 4.4   | 4.3   | 4.3   | 4.7   | 4.8   | 5.8   | 6.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.498 | 0.555 | 0.691 | 0.675 | 0.648 | 0.624 | 0.554 | 0.514 | 0.521 | 0.541 | 0.487 | 0.464 |
|                                                                        | taud        |        | 1.963 | 1.818 | 1.567 | 1.663 | 1.724 | 1.807 | 1.999 | 2.118 | 2.072 | 1.993 | 2.037 | 2.065 |
|                                                                        | Ebn at Noon |        | 801   | 772   | 681   | 689   | 695   | 703   | 758   | 799   | 799   | 774   | 805   | 821   |
|                                                                        | Edh at Noon |        | 181   | 217   | 284   | 257   | 237   | 215   | 179   | 161   | 169   | 179   | 166   | 160   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 5.55  | 6.05  | 6.42  | 6.48  | 6.20  | 5.87  | 5.46  | 5.25  | 5.58  | 5.80  | 5.66  | 5.45  |
|                                                                        | RadStd      |        | 0.31  | 0.29  | 0.27  | 0.28  | 0.27  | 0.19  | 0.25  | 0.22  | 0.25  | 0.14  | 0.20  | 0.19  |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only          | N/A   | N/A     | N/A    | +0.30   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | +0.30   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page